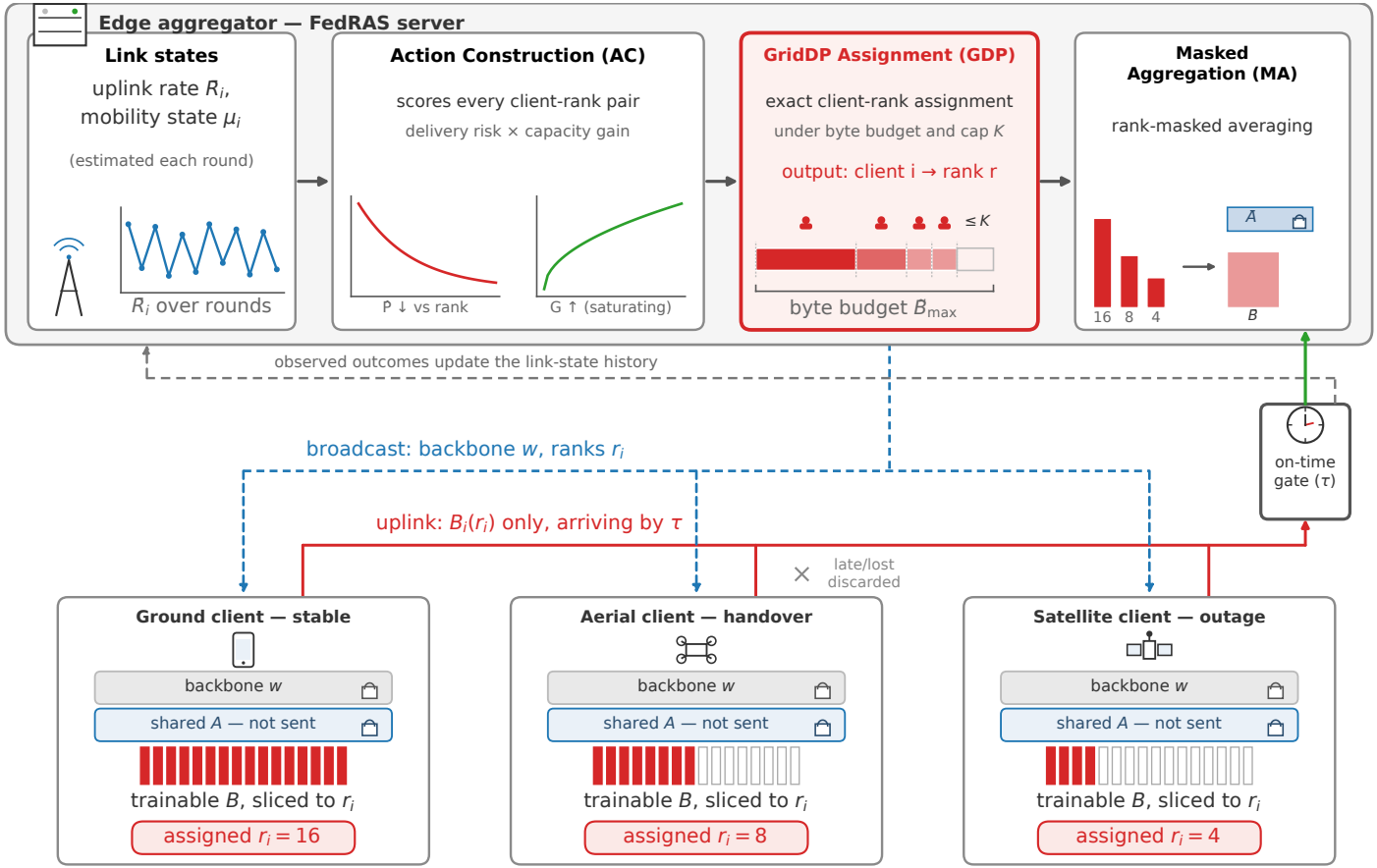
Edge aggregator — FedRAS server
Link states
uplink rate R_i,
mobility state μ_i
(estimated each round)
R_i over rounds
Action Construction (AC)
scores every client-rank pair
delivery risk × capacity gain
P ↓ vs rank
G ↑ (saturating)
GridDP Assignment (GDP)
exact client-rank assignment
under byte budget and cap K
output: client i → rank r
≤ K
byte budget B_{max}
Masked Aggregation (MA)
rank-masked averaging
16 8 4
A
B
observed outcomes update the link-state history
broadcast: backbone w, ranks r_i
uplink: $B_i(r_i)$ only, arriving by τ
on-time gate (τ)
late/lost discarded
Ground client — stable
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 16$
Aerial client — handover
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 8$
Satellite client — outage
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 4$